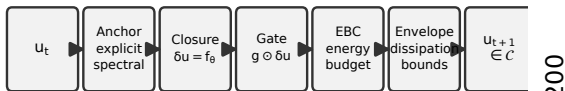


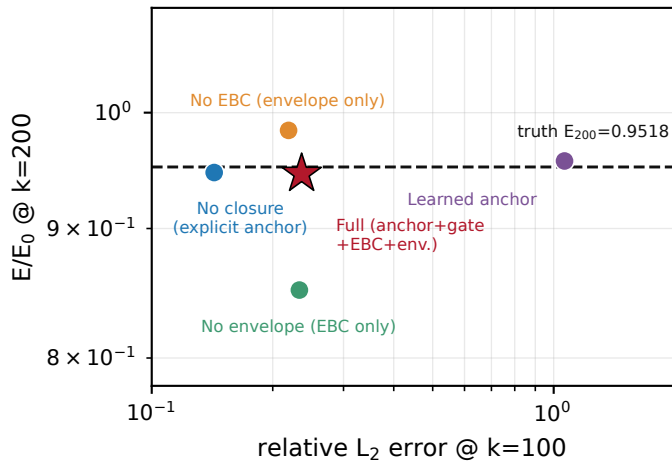
(a) One fused step (v8): Anchor $\rightarrow$ Gate $\rightarrow$ EBC $\rightarrow$ Envelope**Guaranteed per step:**

$$\|u_{\text{fused}} - u_{\text{anc}}\| \leq R_{\text{eff}}(u_t)$$

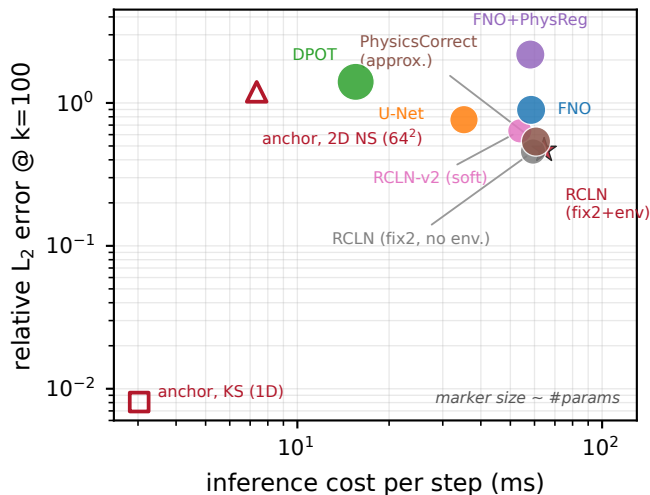
$$E_{\text{in}} - c \varepsilon(u_t) \Delta t \leq E(u_{t+1}) \leq E_{\text{in}}$$

$$\nabla \cdot u_{t+1} = 0 \quad (\text{spectral projection, exact})$$

(b) Component ablation (3D TGV, v8)



(c) Accuracy-cost Pareto (earlier learned-anchor lineage)

(d) Vorticity PDF @ $k=200$: false vs. true stability